

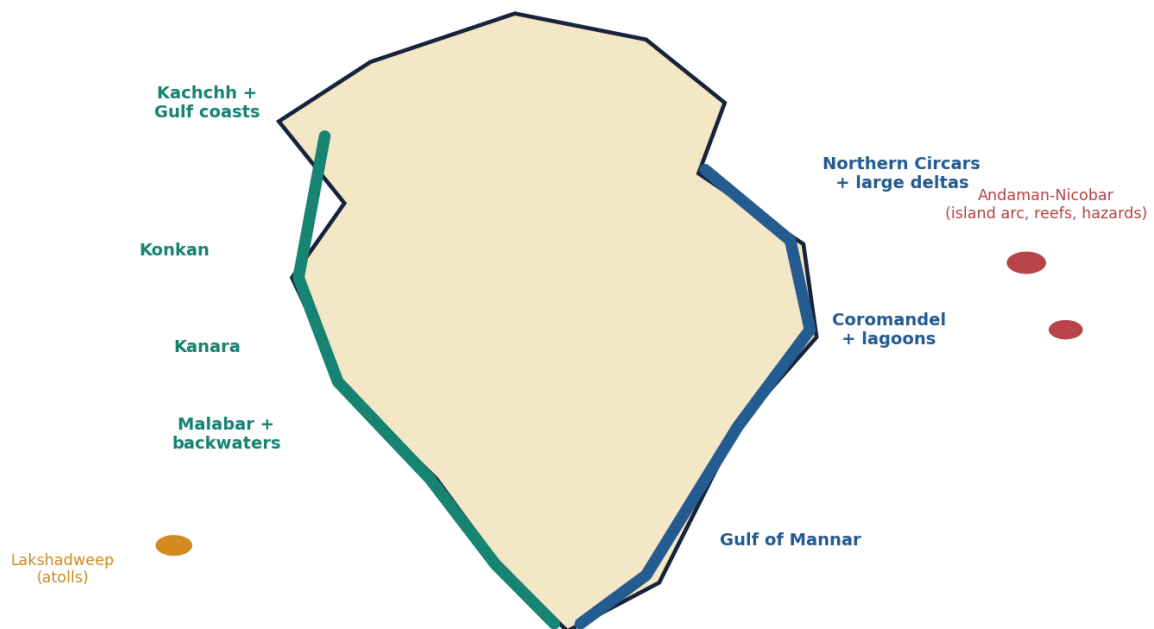
SOLVED PRACTICE WORKBOOK

Geography 10 - India Coast and CRZ - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

India coastal physiography: map-ready schematic

Not to scale. Use bounded examples and process contrasts, rather than a single west-east stereotype.
Gujarat: tidal flats, gulfs, creeks | West: narrower in many sectors | East: broader deltaic sectors



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

Geography 10 - India Coast and CRZ - Complete Topic Package

Solved topic-specific MCQs

Rotation rule: The 56 original package MCQs follow one continuous A -> B -> C -> D sequence. Verified UPSC PYQs retain historical option order and are excluded because a PYQ key cannot be rearranged.

Verified UPSC Prelims PYQs

PYQ-P1 - UPSC Prelims 2018 - coral reefs

Question. Consider the following statements:

1. Most of the world's coral reefs are in tropical waters.
2. More than one-third of the world's coral reefs are located in the territories of Australia, Indonesia and Philippines.
3. Coral reefs host far more number of animal phyla than those hosted by tropical rainforests.

A. 1 and 2 only B. 3 only C. 1 and 3 only D. 1, 2 and 3

Answer: D - INFERRED ANSWER, NOT OFFICIALLY VERIFIED IN THE LOCAL KEY SET. Confidence: high.

- Statement 1: correct; reef-building corals concentrate in warm, shallow, sunlit waters.
- Statement 2: correct in the UPSC framing.
- Statement 3: correct; reefs contain exceptionally high animal-phyla diversity.

PYQ-P2 - UPSC Prelims 2022 - Biorock

Question. "Biorock technology" is talked about in which one of the following situations?

A. Restoration of damaged coral reefs B. Building material from plant residues C. Shale-gas exploration D. Salt licks for wild animals

Answer: A - INFERRED ANSWER, NOT OFFICIALLY VERIFIED IN THE LOCAL KEY SET. Confidence: high.

Low-voltage mineral accretion encourages calcium-carbonate deposition and coral attachment. It does not remove thermal, sewage or sediment stress.

PYQ-P3 - UPSC Prelims 2023 - repeated sea-level falls

Question. Which is the best example of repeated falls in sea level, giving rise to present-day extensive marshland?

A. Bhitarkanika Mangroves B. Marakkanam Salt Pans C. Naupada Swamp D. Rann of Kutch

Answer: D - INFERRED ANSWER, NOT OFFICIALLY VERIFIED IN THE LOCAL KEY SET. Confidence: high.

Rann of Kutch is the intended emerged/marshy former marine setting. Relative sea level and geomorphic history matter; do not force one global cause onto every feature.

PYQ-P4 - UPSC Prelims 2026 - mangrove climate resilience

Question. Which best explain(s) the rationale for protecting mangroves in climate resilience?

1. They reduce tidal energy and store freshwater, making saline estuarine belts ideal for paddy.
2. Salt-sensitive roots filter seawater and convert coast into freshwater aquaculture.
3. By withstanding tidal surges and offering biomass resources, they act as bio-shields and livelihood bases.

A. 1 only B. 1 and 2 C. 2 and 3 D. 3 only

Answer: D - PROVISIONAL LOCAL UPSC KEY, Series A, from Ans-2026-GS1-Provisional.pdf.

Statements 1 and 2 wrongly convert a salt-tolerant tidal ecosystem into a freshwater-production system. Statement 3 is correct with the caveat that protection has thresholds.

Original hard MCQ loop - questions 1-40

MCQ-01

Which best distinguishes a shoreline from CRZ HTL?

- A. Shoreline is an instantaneous water-land intersection; CRZ HTL is an officially demarcated regulatory reference. B. Both are fixed cadastral lines. C. Shoreline is always landward of HTL. D. HTL is any observed wet mark.

Answer: A

Geomorphic position moves over tides/seasons; the notification requires formal HTL demarcation.

MCQ-02

Obliquely breaking waves most directly create:

- A. Permanent mean-sea-level rise. B. Longshore current and zig-zag sediment movement. C. Tidal bores in every estuary. D. Offshore transport only.

Answer: B

Oblique swash plus downslope backwash generates net alongshore transfer.

MCQ-03

A rip current is:

- A. A tide-generating deep-ocean current. B. The same as longshore drift. C. A narrow, fast seaward return flow through the surf zone. D. A wave that always drags swimmers to the seabed.

Answer: C

Water piled nearshore returns through channels; moving parallel to shore is safer than fighting it directly.

MCQ-04

Chronic erosion in a sediment cell is most consistent with:

- A. One post-storm narrowing. B. Inputs persistently exceeding outputs. C. Rising dune volume. D. Outputs persistently exceeding inputs over comparable periods.

Answer: D

Trend diagnosis requires multi-epoch storage and cell-boundary evidence.

MCQ-05

Which idealised erosional sequence is defensible?

- A. Notch/cave -> arch -> stack -> stump. B. Spit -> bar -> lagoon -> delta. C. Dune -> estuary -> arch -> reef. D. Tombolo -> cave -> marsh -> stack.

Answer: A

It is a teaching sequence, not an inevitable path on every lithology.

MCQ-06

A tombolo is:

- A. A drowned river valley. B. A deposit linking an island with mainland or another island. C. A cliff-foot bench. D. A tidal channel through a delta.

Answer: B

Wave refraction and a wave shadow can favour deposition between land masses.

MCQ-07

Relative sea-level rise at a delta may result from:

- A. Global sea rise only. B. Tectonic uplift only. C. Ocean rise plus subsidence/compaction, with sediment partly offsetting local effects. D. Annual rainfall alone.

Answer: C

Relative level compares water and land; sediment budget shapes the shoreline response.

MCQ-08

Which statement about India's west/east coasts is best?

- A. Entire west is erosional and east depositional. B. Only east has estuaries. C. Arabian Sea cyclones are impossible. D. Broad contrasts help, but Gujarat gulfs, rocky sectors, deltas, lagoons and engineering create exceptions.

Answer: D

UPSC rewards qualified regional differentiation, not rigid binaries.

MCQ-09

Which pair is correctly matched?

- A. Narmada-Tapi: estuarine mouths influenced by tides. B. Godavari-Krishna: classic rias. C. Chilika: ocean trench. D. Vembanad: coral atoll.

Answer: A

Narmada/Tapi enter the high-tidal Gulf of Khambat; major east-flowing rivers commonly form deltas.

MCQ-10

A coastal lagoon is:

- A. Any oxbow near a coast. B. A partly enclosed water body behind a barrier/spit with inlet exchange. C. Water beyond 12 nautical miles. D. A permanently freshwater aquifer.

Answer: B

Inlet position, salinity and exchange can vary seasonally.

MCQ-11

Best evidence separating chronic erosion from seasonal variability:

- A. One post-cyclone photograph. B. Presence of a seawall. C. Comparable multi-epoch shoreline and beach-volume records. D. One undated interview.

Answer: C

Time-consistent morphology is necessary to diagnose persistent loss.

MCQ-12

A harbour arm across active littoral drift can:

- A. Guarantee equal accretion both sides. B. Stop relative sea-level rise. C. Eliminate monitoring. D. Accrete sand updrift and create downdrift deficit without bypassing/nourishment.

Answer: D

The structure redistributes a finite sediment flux; exact response is site-specific.

MCQ-13

Storm surge is:

- A. Abnormal coastal water-level rise driven mainly by cyclone winds/pressure, modified by tide, track and bathymetry. B. Astronomical tide alone. C. A fault-generated tsunami. D. Permanent mean-sea-level rise.

Answer: A

Surge, tide, waves and rainfall can combine into compound inundation.

MCQ-14

Which belongs to a tsunami-warning chain?

A. Rain gauges only. B. Seismic networks, bottom-pressure recorders, tide gauges, models and communication. C. Coastal CCTV alone. D. Television only after landfall.

Answer: B

Detection and confirmation must connect to actionable dissemination and evacuation.

MCQ-15

Coastal disaster risk is best represented by:

A. Wave height alone. B. Population alone. C. Hazard interacting with exposure and vulnerability, modified by capacity. D. Shelter count alone.

Answer: C

The same surge produces different losses in differently exposed and prepared communities.

MCQ-16

Soundest statement about mangrove protection:

A. Mangroves stop every tsunami. B. They work equally regardless of width/condition. C. Planting guarantees a functioning ecosystem. D. They can attenuate selected flows and support livelihoods, but cannot replace evacuation everywhere.

Answer: D

Hydrology, forest structure, water depth and event intensity matter.

MCQ-17

CRZ along the sea front under the 2019 base includes:

A. HTL to 500 m landward plus specified intertidal/water areas. B. Only HTL-LTL. C. Every coastal-district parcel. D. Only ports.

Answer: A

CRZ includes land, intertidal and water categories; classification and activity govern regulation.

MCQ-18

For CRZ 2019, HTL means:

A. Any storm-debris line. B. Highest water reached during spring tide, demarcated under NCSCM procedure. C. The national 1:250,000 coastline in every case. D. Low-water line on an ENC.

Answer: B

Official demarcation, not visual guesswork, governs the notification.

MCQ-19

Which belongs to CRZ-I A?

A. Every municipal ward within 500 m. B. All water beyond territorial sea. C. Biologically active mudflats. D. Every cultivated coastal field.

Answer: C

Other examples include mangroves, reefs, dunes, salt marshes and seagrass.

MCQ-20

CRZ-I B is:

A. All land 500 m from HTL. B. Only mangrove buffers. C. All developed urban land. D. Intertidal area between LTL and HTL.

Answer: D

It is distinct from CRZ-I A ecologically sensitive areas.

MCQ-21

CRZ-II requires:

A. Legally urban area, substantial built-up condition and specified infrastructure. B. Any village with a beach. C. Density over 2161/km² alone. D. Any port.

Answer: A

The built-up and infrastructure test matters; “urban” alone is insufficient.

MCQ-22

CRZ-III A uses:

A. Density below 2161/km² with automatic 50 m NDZ. B. Density above 2161/km² on 2011 Census base, with 50 m NDZ after approved updated CZMP. C. No density criterion. D. A 20 m rule for all mainland villages.

Answer: B

Without updated-plan approval, the 200 m NDZ continues for this relaxation.

MCQ-23

CRZ-III B generally has:

A. Automatic CRZ-II classification. B. No landward regulation. C. Density below 2161/km² and 200 m NDZ. D. Only water and seabed.

Answer: C

The actual category must still be read from the approved plan.

MCQ-24

CRZ-IV B covers:

A. Landward dunes. B. Municipal construction. C. Offshore islands only. D. Water and bed of tidal-influenced bodies to defined tidal influence.

Answer: D

The 5 ppt salinity criterion in the driest season helps define tidal influence.

MCQ-25

The 50 m tidal-water-body reach under 2019:

A. Depends on final updated-CZMP approval; meanwhile 100 m/creek-width continues. B. Applied automatically nationwide in January 2019. C. Never applies to creeks. D. Is a coastline-length rule.

Answer: A

This is a central transition caveat.

MCQ-26

The CRZ hazard line is best used as:

A. A universal prohibition identical to NDZ. B. A disaster-management/adaptation input for land-use planning. C. A substitute for HTL. D. A maritime boundary.

Answer: B

The notification describes it as a planning tool, not a one-size setback.

MCQ-27

Why did 2011 procedures continue in some places after 2019 publication?

A. The 2019 Gazette had no legal force. B. States can ignore central rules indefinitely. C. Operational changes are linked to approved updated CZMPs, so transition is plan-specific. D. Only courts can prepare maps.

Answer: C

Notification, plan approval and project appraisal are distinct stages.

MCQ-28

A non-EIA project purely in CRZ-II/III is generally considered for CRZ clearance by:

A. NCCR. B. Survey of India. C. INCOIS. D. The concerned CZMA.

Answer: D

CRZ-I/IV involvement generally shifts the route to MoEFCC on CZMA recommendation.

MCQ-29

A project attracting CRZ 2019 and EIA Notification 2006 generally receives:

A. Composite environmental and CRZ treatment through the competent EIA authority with CZMA recommendation. B. Only a trade licence. C. Only consent to operate. D. No appraisal.

Answer: A

The exact authority depends on EIA category and CRZ location.

MCQ-30

Which distinction is correct?

A. CRZ clearance automatically grants consent to operate. B. Pollution-control consent is distinct from CRZ/EC clearance and compliance. C. Consent proves zero impact. D. CZMP approval is project consent.

Answer: B

Different laws and conditions govern different stages.

MCQ-31

As of 16 August 2026, G.S.R. 410(E), 26 May 2026 was:

A. The final 2019 notification. B. A Supreme Court order. C. A draft proposing specified chemical storage outside CRZ-I A. D. A national CZMP approval.

Answer: C

The "Draft Notification" heading and objection window determine status.

MCQ-32

The 2025 increase in measured coastline is:

A. Rapid land creation. B. A substitute for CRZ HTL. C. Proof every beach accreted. D. A remeasurement effect from scale, method, HWL treatment and island inclusion.

Answer: D

Measurement conventions can change length without changing sovereignty.

MCQ-33

The Odisha cyclone lesson is:

A. Forecast-linked evacuation, shelters, drills and resilient administration, supported but not replaced by ecosystems. B. Mangroves alone eliminate risk. C. Warnings matter only after landfall. D. Hard structures make evacuation unnecessary.

Answer: A

Layered risk reduction is more transferable than a single-cause story.

MCQ-34

The Puducherry restoration lesson illustrates:

A. Every seawall creates a permanent beach. B. Restoring sediment transfer/nourishing a harbour-affected coast can rebuild beach but needs monitoring and supply. C. Nourishment never needs replenishment. D. Harbours do not affect

drift.

Answer: B

The mechanism is sediment-budget correction, not cosmetic sand placement.

MCQ-35

Prudent intervention hierarchy:

A. Seawalls everywhere before diagnosis. B. Plant mangroves after blocking tidal flow. C. Avoid -> protect processes -> accommodate -> restore/hybrid -> justified hard defence -> context-specific retreat. D. Approve development before hazard mapping.

Answer: C

The hierarchy prevents high-cost lock-in.

MCQ-36

Best monitoring design for a groyne field:

A. Completion certificate only. B. One updrift photograph. C. Original clearance letter only. D. Before-after and updrift-downdrift volumes, bypass records and triggers.

Answer: D

This measures redistribution, performance and unintended effects.

MCQ-37

Mangrove restoration is most likely to succeed when:

A. Tidal hydrology, elevation and sediment conditions are restored with suitable species recovery. B. Any species is planted in rows on any mudflat. C. The site is isolated from tides. D. One-month survival equals full recovery.

Answer: A

Hydrology is often the first-order constraint.

MCQ-38

Beach nourishment is:

A. A permanent one-time cure. B. Adding compatible sediment with borrow, replenishment and monitoring obligations. C. A substitute for cell analysis. D. Concrete armouring.

Answer: B

It works with the beach but remains a maintained intervention.

MCQ-39

Managed retreat means:

A. Unplanned abandonment after every storm. B. A mandatory rule for every coast. C. Planned relocation/rolling adaptation where residual risk makes permanent defence unsuitable. D. Prohibition of fishing.

Answer: C

It requires rights, finance and livelihood safeguards.

MCQ-40

Which statement about clearance and outcome is correct?

A. Clearance proves no future erosion. B. Any erosion proves illegality. C. Consent substitutes for monitoring. D. Clearance authorises subject to conditions; success requires compliance and outcome indicators.

Answer: D

Lawful permission, compliance and environmental outcome are separate empirical questions.

Remedial MCQ loop - questions 41-56

The key rotation continues without reset.

R-MCQ-41

A beach narrows during monsoon and rebuilds later. Safest conclusion:

A. Seasonal variability unless multi-epoch volume evidence shows chronic loss. B. Permanent erosion is proven. C. Sea-level rise is absent. D. Annual accretion is proven.

Answer: A

Do not infer trend from one phase of a seasonal cycle.

R-MCQ-42

Most accurate statement about seawalls:

A. They always cause identical downdrift erosion. B. They may defend hinterland locally but narrow beach/alter adjacent process; effect is site-specific. C. They always rebuild recreational beach. D. They remove surge risk.

Answer: B

Separate asset protection from beach and cell outcomes.

R-MCQ-43

A delta sector accretes despite regional relative sea-level rise. This means:

A. Sea-level data are false. B. Subsidence is impossible. C. Local deposition can temporarily exceed accommodation demand. D. Every delta sector is safe.

Answer: C

Relative water level and sediment budget jointly shape shoreline position.

R-MCQ-44

A table says CRZ-IIIA has 50 m NDZ everywhere. Correction:

A. IIIA has no NDZ. B. 50 m applies beyond territorial waters. C. States can waive it informally. D. 50 m requires density criterion and approved updated CZMP; otherwise 200 m continues.

Answer: D

Every CRZ number must travel with its condition.

R-MCQ-45

NCZMA recommends a CZMP. Immediate conclusion:

A. Recommendation is not final MoEFCC approval; check the later order. B. Every project is automatically cleared. C. The plan is rejected. D. CRZ ceases to apply.

Answer: A

The Daman-Diu 2026 sequence demonstrates the distinction.

R-MCQ-46

An approved 1:25,000 CZMP means a parcel project still needs:

A. No review. B. Applicable local/project mapping and appraisal/clearance/consent. C. Only coastline-length data. D. A newspaper report.

Answer: B

Plan approval enables regulation; it does not approve each project.

R-MCQ-47

The 11,098.81 km coastline figure should be used in a CRZ dispute as:

A. Proof HTL moved seaward. B. Proof a beach accreted. C. A national measurement fact distinct from parcel regulatory HTL/CZMP. D. Substitute for approved maps.

Answer: C

Purpose and scale control evidentiary use.

R-MCQ-48

A 2026 draft proposes additional chemical storage. A Mains answer should say:

A. It became final on publication. B. All CRZ-I A storage became legal. C. It repealed CRZ 2019. D. It remained draft on the retrieval date unless a later final Gazette is verified.

Answer: D

Status words are legally substantive.

R-MCQ-49

A mangrove plantation has 80% one-year survival. Strongest evaluation:

A. Promising establishment; hydrology, recruitment, habitat and livelihood outcomes still need tracking. B. Full restoration is proven. C. Surge risk is eliminated. D. Monitoring can end.

Answer: A

Output and outcome must be separated.

R-MCQ-50

A nourishment project widens a beach immediately. Next priority:

A. Declare permanent success. B. Track retention, downdrift effects, borrow impacts and replenishment triggers. C. Stop profiles. D. Build on the new berm.

Answer: B

Dynamic interventions need maintenance criteria.

R-MCQ-51

A cyclone warning is accurate but evacuation fails. Missing concept:

A. Wave refraction. B. Territorial-sea delimitation. C. End-to-end warning: trust, access, routes, transport, shelters and action. D. Coral mineral accretion.

Answer: C

Forecast skill does not automatically become avoided loss.

R-MCQ-52

A port EIA ignores nearby ports and river-sediment change. Key gap:

A. Tourism plan only. B. Coastline-length calculation. C. Coral definition. D. Cumulative and sediment-cell assessment.

Answer: D

Multiple interventions interact across the same littoral system.

R-MCQ-53

Improve “west coast = estuaries, east coast = deltas” by:

A. Add Gujarat gulfs and local exceptions, east-coast lagoons/rocky sectors and mixed processes. B. Remove all comparison. C. Say west has no deposition. D. Say east has no tides.

Answer: A

Qualified generalisation earns more than a rigid binary.

R-MCQ-54

Best first response to chronic dune erosion beside a new resort:

- A. Flatten the dune. B. Restore access/sediment pathways, reassess setback and diagnose beach-dune budget. C. Plant mangroves on dry crest. D. Assume offshore currents alone.

Answer: B

Match intervention to landform and causal pathway.

R-MCQ-55

Pollution-control consent proves:

- A. Approved CZMP. B. Acceptable cumulative impacts. C. A distinct pollution authorisation; CRZ/EC and compliance remain separate checks. D. No future violation.

Answer: C

Permissions are law-specific and conditional.

R-MCQ-56

Strongest final coastal-management verdict:

- A. Ban all infrastructure. B. Armour the entire shore. C. Depend only on ecosystems. D. Use adaptive, cell-based and equitable management with measurable triggers.

Answer: D

A ranked portfolio can match different assets, processes and residual risks.

Verified UPSC Mains PYQs with model solutions

M-PYQ 1 - UPSC GS-I 2019 - coral life and global warming - 10 marks

Question: Assess the impact of global warming on the coral life system with examples. (150 words)

Model solution

Claim: Global warming damages coral systems mainly through marine heatwaves that disrupt coral-algal symbiosis, causing bleaching, mortality and reef-structure loss. **Named evidence/example:** Repeated bleaching across the Great Barrier Reef and thermal stress in Indian reef regions such as Lakshadweep and Gulf of Mannar show heat combining with pollution and sediment stress. **Analysis:** Mortality reduces habitat complexity, nursery grounds, fisheries and tourism; weaker reef crests also reduce offshore wave dissipation. Acidification lowers calcification potential. **Qualification:** Bleaching is not immediate death; recovery depends on duration, recurrence, tolerance, recruitment and local water quality. Local action cannot stop global heating but can improve resilience by controlling sewage, destructive fishing, anchors and sedimentation. **Verdict:** Combine rapid global mitigation with enforced water-quality, fisheries, restoration and heat-response plans.

Why this earns marks: It links climate forcing to biology, names Indian/global evidence, adds service impacts, qualifies bleaching and ends with a two-scale verdict.

M-PYQ 2 - UPSC GS-I 2019 - mangrove depletion and importance - 10 marks

Question: Discuss the causes of depletion of mangroves and explain their importance in maintaining coastal ecology. (150 words)

Model solution

Claim: Mangroves decline when tidal hydrology, sediment supply and habitat space are disrupted, not merely when trees are cut. **Named evidence/example:** Reclamation and port/urban pressure around estuaries, aquaculture conversion in deltas and embankment-channel change in the Sundarbans illustrate pathways; Bhitarkanika shows nursery and hazard-buffer functions. **Analysis:** Mangroves stabilise selected sediments, cycle nutrients, store carbon, support fish/crustacean nurseries and provide livelihoods. Root-canopy structure can reduce flow and wave energy. **Qualification:** Performance depends on width, condition, bathymetry and event magnitude; a narrow plantation cannot replace evacuation. Planting without tidal exchange may fail. **Verdict:** Protect existing hydrologically

connected forests first, restore tidal processes second and monitor ecology plus community access.

Why this earns marks: It moves beyond plantation rhetoric, uses cases, explains mechanisms, gives a threshold caveat and prioritises process restoration.

M-PYQ 3 - UPSC GS-III 2019 - coastal sand mining - 10 marks

Question: Coastal sand mining, whether legal or illegal, poses one of the biggest threats to our environment. Analyse the impact of sand mining along Indian coasts, citing specific examples. (150 words)

Model solution

Claim: Mining removes sediment from a finite beach-dune-river system and can convert a balanced littoral budget into chronic deficit. **Named evidence/example:** Mining pressure in Kerala/Goa coastal and river-linked systems and dune removal highlighted after the 2004 tsunami illustrate exposure; placer-rich Tamil Nadu/Odisha/Kerala tracts show the need to distinguish regulated atomic-mineral activity from illegal bulk removal. **Analysis:** Effects include beach lowering, dune weakening, saline intrusion, habitat loss, turbidity, infrastructure exposure and reduced storm recovery. River mining can reduce downstream coastal supply. **Qualification:** Impact depends on volume, grain size, extraction location, replenishment and cell connectivity; not every permitted removal has equal effect. **Verdict:** Use sediment budgets, transparent permits, remote-sensing plus field enforcement, no-mining active stores and restoration liability.

Why this earns marks: It supplies the sediment mechanism, Indian contexts, ecological/hazard effects, legal nuance and measurable enforcement.

M-PYQ 4 - UPSC GS-III 2022 - coastal erosion - 15 marks

Question: Explain the causes and effects of coastal erosion in India. What management techniques are available for combating the hazard? (250 words)

Model solution

Claim: Coastal erosion is persistent sediment-volume or shoreline loss caused by interacting wave, sediment, land-level and human pressures; seasonal monsoon narrowing alone is not chronic erosion. **Named evidence/example:** NCCR classified about 33.6% of assessed mainland shoreline as eroding during 1990-2018. Harbour-linked interruption and restoration at Puducherry, delta vulnerability in Sundarbans and dense Kerala/Goa occupation show different mechanisms. **Analysis:** Natural drivers include storms, inlet migration, wave variability and relative sea-level rise. Human drivers include sediment trapping by dams, sand mining, ports/groynes, dredging, dune loss, reclamation and subsidence. Effects include land/infrastructure loss, salinity, habitat decline, displacement and repeated defence expenditure. **Techniques:** Avoid-risk zoning; protect dunes/tidal channels and river sediment; nourish beaches; restore dunes/mangroves; bypass sediment at ports; use site-justified seawalls/revetments/groynes; strengthen warning/accommodation; consider managed retreat where residual risk is intolerable. **Qualification:** Every intervention redistributes benefits and maintenance costs. A seawall may defend land yet narrow beach; ecosystems have thresholds. **Verdict:** Use sediment-cell plans with before-after/updrift-downdrift monitoring, access safeguards and adaptive triggers.

Why this earns marks: It defines erosion, uses dated official evidence, classifies causes/effects, ranks solutions and qualifies hard and nature-based measures.

M-PYQ 5 - UPSC GS-I 2023 - resource potential and hazard preparedness - 15 marks

Question: Comment on the resource potentials of the long coastline of India and highlight the status of natural hazard preparedness in these areas. (250 words)

Model solution

Claim: India's coast is a production, connectivity and ecological frontier, but resource value is sustainable only when compound hazard risk is internalised. **Named evidence/example:** Ports/trade, fisheries, offshore energy, tourism, salt/minerals, mangrove blue carbon and reefs/seagrass show diversity. The 2025 official remeasurement records 11,098.81 km through changed method, not new territory. Odisha evacuation/shelters and INCOIS warning systems show preparedness capacity. **Analysis:** Cyclones, surge, tsunami, erosion, flooding, saline intrusion and rip currents threaten dense settlements and infrastructure. India has IMD/INCOIS detection, NDMA/state plans, shelters, evacuation practice, CRZ/CZMP tools and restoration. Yet last-mile trust, informal housing, local maps, maintenance, cumulative urban/port effects and enforcement remain uneven. **Qualification:** Preparedness cannot be judged only

by forecast accuracy or clearance count. Outcomes depend on accessible routes, functioning shelters, compliant land use and ecosystem condition. **Verdict:** Convert blue-economy planning into risk-screened ICZM with interoperable warning, approved/local CZMPs, cell monitoring and livelihood access.

Why this earns marks: It balances resources and risks, uses current official evidence, separates capacity from outcome and gives an integrated verdict.

M-PYQ 6 - UPSC GS-I 2025 - tsunami - 10 marks

Question: What are Tsunamis? How and where are they formed? What are their consequences? Explain with examples. (150 words)

Model solution

Claim: A tsunami is a series of long-period waves generated by rapid displacement of a large water volume, commonly from vertical seabed movement in a major undersea earthquake. **Named evidence/example:** The 26 December 2004 Indian Ocean event near the Sunda subduction zone caused severe inundation in Tamil Nadu, Andhra Pradesh, Kerala and Andaman-Nicobar. Landslides and volcanic collapse can also generate tsunamis.

Analysis: Offshore, low steepness enables fast travel; near shore, slowing and shoaling raise height and currents. Consequences include inundation, debris impact, erosion, salinisation, port damage, habitat loss and cascading utility failures. **Qualification:** Not every submarine earthquake produces a tsunami; magnitude, depth, fault motion and displacement matter. Ecosystems cannot replace warning and evacuation. **Verdict:** Combine INCOIS detection/modelling with last-mile alerts, mapped routes, drills, evacuation options and risk-sensitive land use.

Why this earns marks: It defines, explains generation/shoaling, names an Indian example, covers cascading effects, qualifies causation and gives an end-to-end response.

Original solved Mains practice

Practice 1 - 10 marks - CRZ implementation

Question: "CRZ Notification 2019 did not create an identical regulatory switch along every Indian coast on publication." Explain. (150 words)

Model solution

Claim: The notification changed national law, but several operational provisions depend on state/UT-specific approved updated CZMPs. **Named evidence/example:** The 50 m tidal-water-body reach and 50 m CRZ-IIIA NDZ apply after plan approval; otherwise specified 2011 distances continue. NCZMA recommended Daman-Diu on 8 May 2026, followed by separate final MoEFCC approval on 18 June. **Analysis:** Implementation requires NCSCM HTL/LTL inputs, 1:25,000 plans, consultation, CZMA appraisal, ministry approval, project mapping and clearance. Notification, recommendation, approval and project permission are distinct. **Qualification:** Plan approval also does not legalise each project; EIA, pollution consent, planning permission and compliance may remain necessary. **Verdict:** Current CRZ answers must cite the base notification plus relevant plan/order and retrieval date.

Why this earns marks: It uses conditional distances, a 2026 document sequence, institutional logic and a precise legal verdict.

Practice 2 - 15 marks - sediment cells and structures

Question: Analyse why coastal structures should be appraised at sediment-cell scale rather than only at the protected frontage. (250 words)

Model solution

Claim: Beaches are connected stores; a structure can protect one frontage while redistributing risk beyond its boundary. **Named evidence/example:** Harbour arms can trap drift updrift and create downdrift deficit, illustrated by the process diagnosis behind Puducherry beach-restoration work. Groynes, breakwaters and dredged channels similarly alter waves, bypassing and inlet deltas. **Analysis:** Frontage-only appraisal misses river supply, seasonal drift reversal, offshore stores, adjacent structures and cumulative effects. A seawall may prevent land retreat but cause reflection/scour and dry-beach loss. Cell appraisal compares sources, transfers, stores and sinks, then tests setback, bypassing, nourishment, dune restoration or targeted hard defence. **Qualification:** No universal rule says every

structure causes the same downdrift erosion. Orientation, transport rate, geology, wave climate, permeability and maintenance matter. **Verdict:** Conditions should require baseline profiles/bathymetry, cumulative modelling, sediment plans, updrift-downdrift monitoring, disclosure and redesign/compensation triggers.

Why this earns marks: It explains mechanism, uses an Indian case, avoids absolutism and converts analysis into enforceable conditions.

Practice 3 - 20 marks - climate-resilient ICZM

Question: Design a climate-resilient and socially just integrated coastal-zone management framework for India. (250 words)

Model solution

Claim: Resilient ICZM must manage interacting shoreline, watershed, ecosystem, settlement and livelihood systems rather than choose between development and conservation. **Named evidence/example:** NCCR's 1990-2018 assessment shows mixed erosion/accretion/stability; Odisha shows warning-evacuation-shelter value; Sundarbans and Puducherry illustrate relative-level and sediment-cell problems; CRZ 2019 supplies plan/clearance architecture. **Analysis:** A six-part framework should: (1) avoid new exposure through hazard-sensitive CZMP/local plans; (2) protect river sediment, dunes, wetlands and access; (3) ensure last-mile IMD/INCOIS evacuation; (4) restore mangroves, seagrass, reefs and dunes where process conditions fit; (5) use nourishment, bypassing and targeted hard structures after cumulative appraisal; (6) fund accommodation/retreat with tenure, compensation and livelihood continuity. Common maps and public compliance dashboards should link MoEFCC/CZMAs, pollution boards, local bodies and disaster authorities. **Qualification:** Ecosystems have thresholds, retreat is distributionally difficult, and hard structures remain justified for some critical assets. **Verdict:** Measure success through beach volume, ecosystem condition, flood loss, compliance, access and livelihoods - not plan count or construction completion.

Why this earns marks: It integrates process, law, institutions, equity and indicators; uses named evidence and a qualified portfolio verdict.